HSE 103 Final Project: R Analysis

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I used AI to help me understand parts of the assignment and fix errors in my R code. I reviewed the results myself.

Introduction

For this project, I analyzed the expression of three genes, *CNR1*, *SOCS5*, and *SIRT3*, across 157 pregnancy samples. I used maternal BMI as my continuous variable and pregnancy status as my first categorical variable. I created a second categorical variable by dividing maternal BMI into two groups, below 25 and 25 or higher. I created an R function to generate a histogram, scatterplot, and grouped boxplot for each gene, giving a total of nine plots. I then compared the gene expression patterns across the three genes.

Gene expression plots

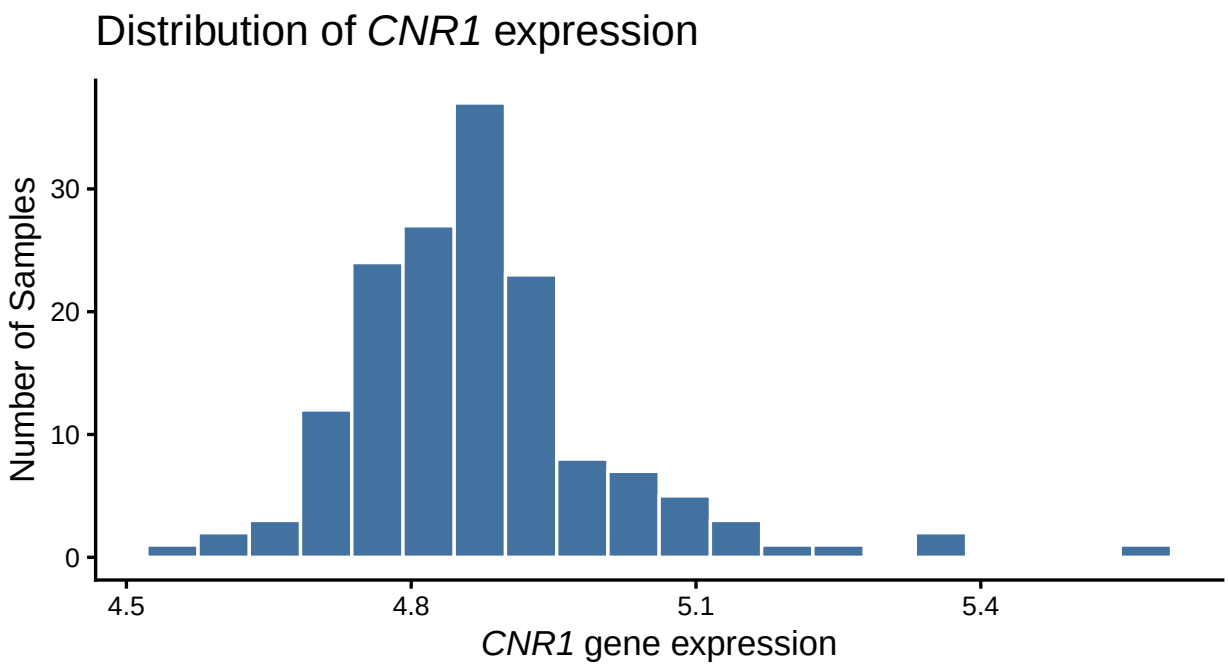


Figure 1: Distribution of CNR1 expression across 157 pregnancy samples.

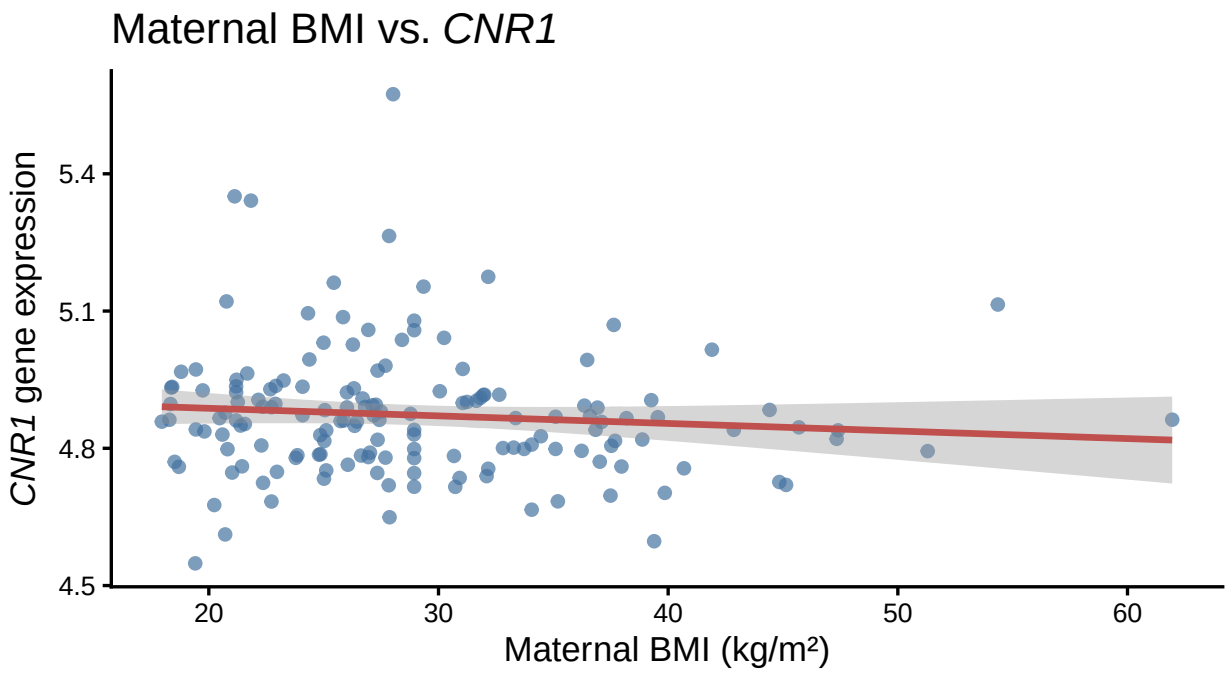


Figure 2: Relationship between maternal BMI and CNR1 expression.

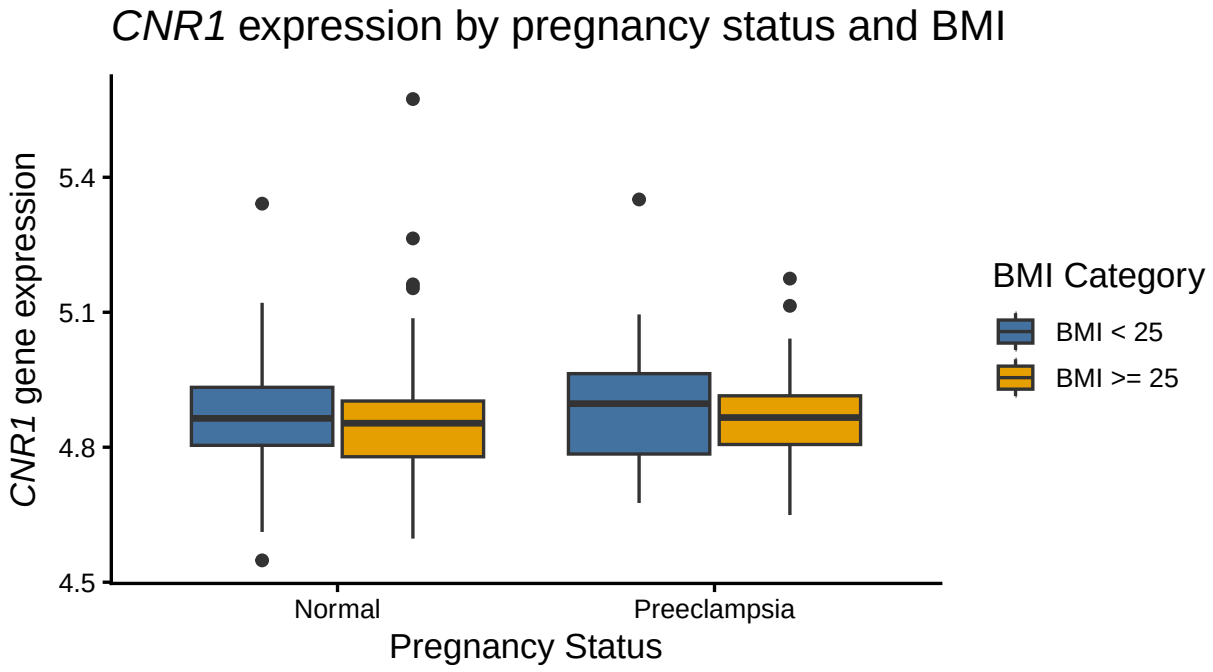


Figure 3: CNR1 expression by pregnancy status and BMI category.

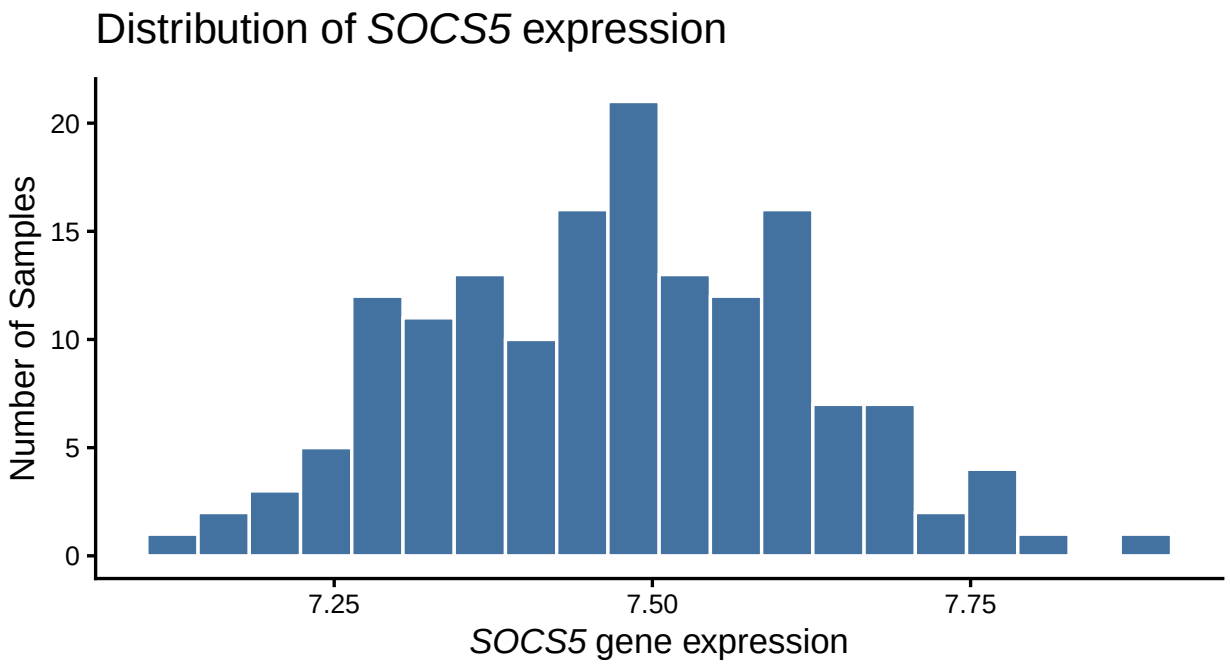


Figure 4: Distribution of SOCS5 expression across 157 pregnancy samples.

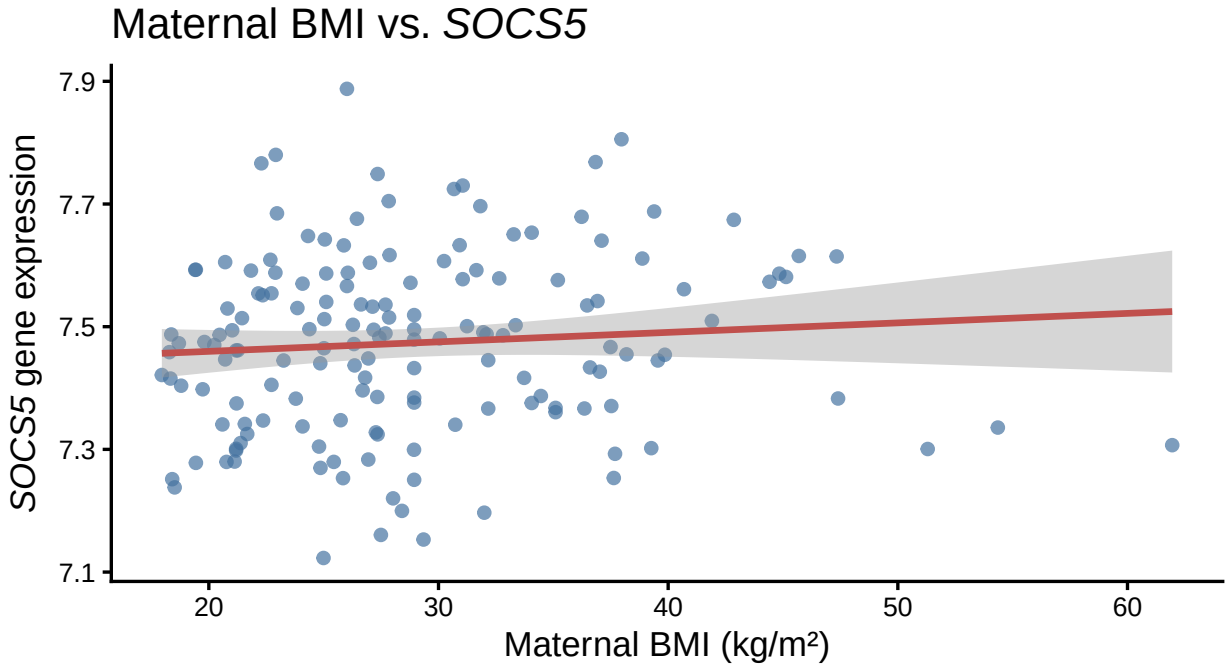


Figure 5: Relationship between maternal BMI and SOCS5 expression.

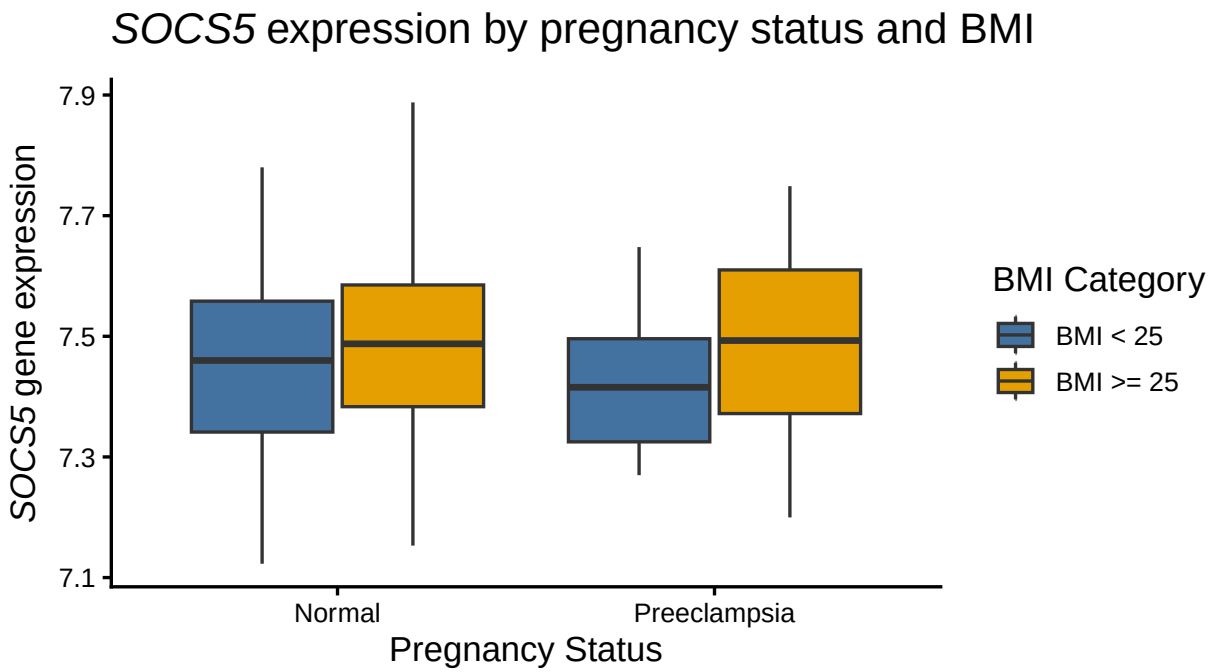


Figure 6: SOCS5 expression by pregnancy status and BMI category.

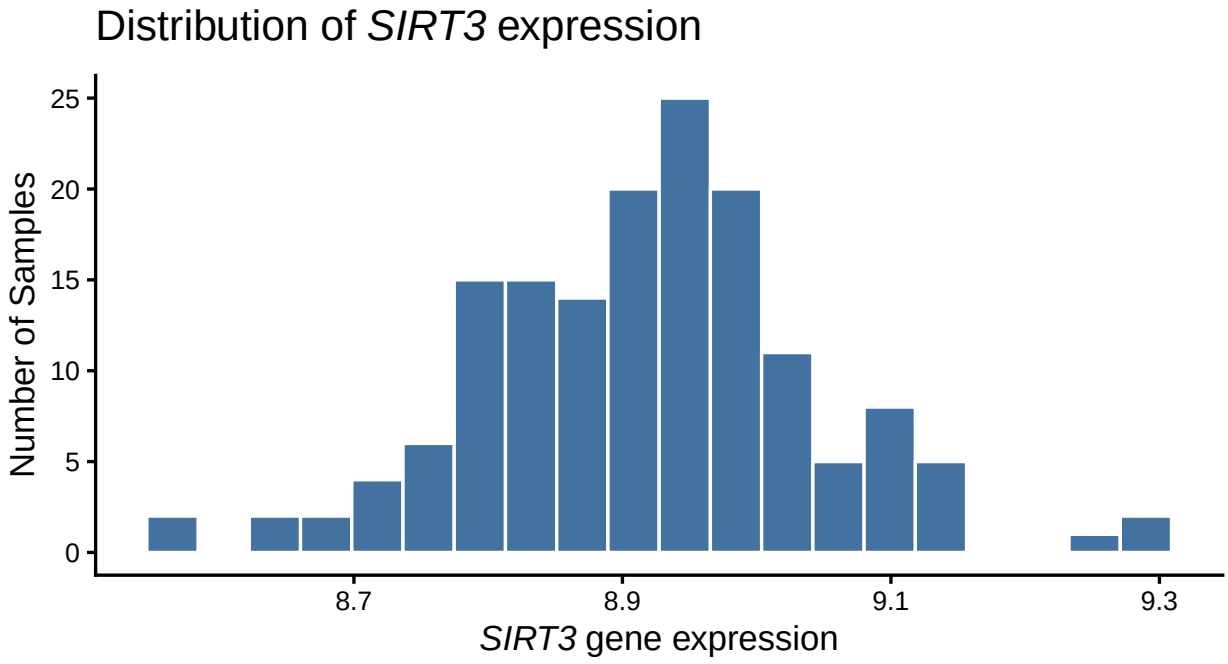


Figure 7: Distribution of SIRT3 expression across 157 pregnancy samples.

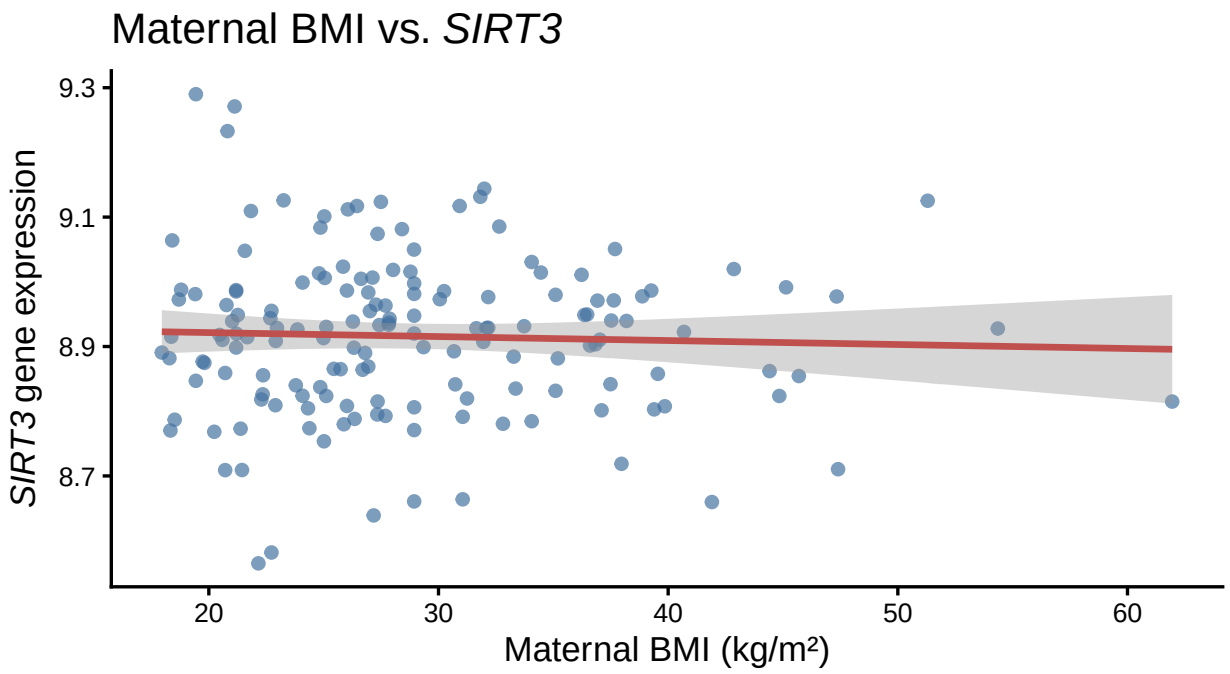


Figure 8: Relationship between maternal BMI and SIRT3 expression.

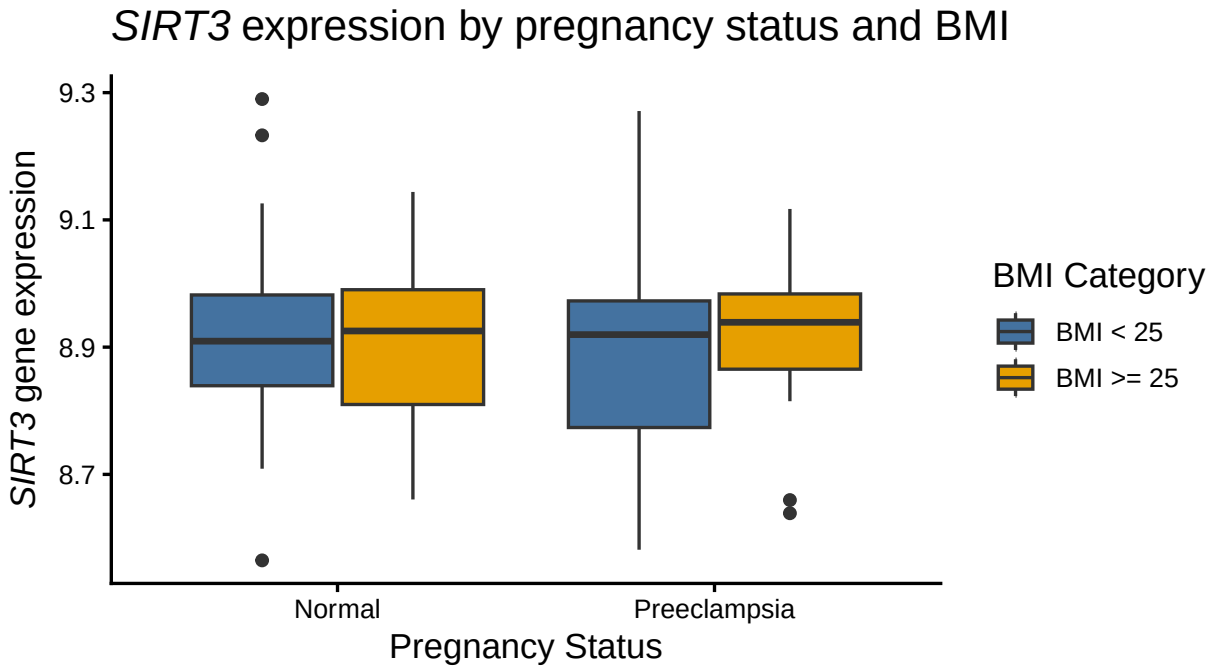


Figure 9: SIRT3 expression by pregnancy status and BMI category.

Table 1: Descriptive statistics for three genes

Gene	Mean	Median	SD	Minimum	Maximum
CNR1	4.873	4.861	0.140	4.549	5.574
SOCS5	7.474	7.479	0.146	7.123	7.888
SIRT3	8.916	8.923	0.123	8.565	9.290

Comparison of Gene Expression Across Three Genes

I compared *CNR1*, *SOCS5*, and *SIRT3* using their mean, median, standard deviation, minimum, and maximum expression values across 157 pregnancy samples.

SIRT3 had the highest average expression value (8.916), followed by *SOCS5* (7.474) and *CNR1* (4.873).

SOCS5 had the highest standard deviation (0.146), suggesting slightly more variation in its expression values. *SIRT3* had the lowest standard deviation (0.123), suggesting slightly less variation.

CNR1 had the largest difference between its minimum and maximum expression values, mainly because a few samples had higher expression values.

Overall, the three genes showed different expression distributions. These results describe expression patterns within the dataset and do not establish biological or statistical significance.

Table 2: Pearson correlations with maternal BMI

Gene	BMI_Correlation
CNR1	-0.093
SOCS5	0.084
SIRT3	-0.039

Comparison of Maternal BMI and Gene Expression

I calculated Pearson correlations to compare the relationship between maternal BMI and the expression of *CNR1*, *SOCS5*, and *SIRT3* across 157 pregnancy samples.

CNR1 had a correlation of -0.093, showing a very weak negative relationship with maternal BMI. *SOCS5* had a correlation of 0.084, showing a very weak positive relationship. *SIRT3* had a correlation of -0.039, which was closest to zero.

Overall, all three correlations were very weak. This suggests that maternal BMI had little linear association with the expression of these genes in this dataset. These results do not establish causation or statistical significance.

Comparison of Gene Expression by Pregnancy Status and BMI

I compared *CNR1*, *SOCS5*, and *SIRT3* expression across normal pregnancies and pregnancies with preeclampsia. I also compared two BMI categories: below 25 and 25 or higher.

For *CNR1*, the median expression values were similar across all four groups. The BMI below 25 groups showed slightly higher median expression.

For *SOCS5*, participants with BMI 25 or higher generally had higher median expression values, especially among those with preeclampsia.

Table 3: **Maternal characteristics by pregnancy status**

Characteristic	**Overall** N = 157[1]	**Normal** N = 110[1]	**Preeclampsia** N = 47[1]
Maternal Age (years)	26.38 (5.04)	26.63 (5.08)	25.78 (4.95)
Maternal BMI	28.94 (7.88)	27.99 (7.66)	31.16 (8.03)
Gestational Age at Delivery (weeks)	38.82 (2.18)	39.25 (1.02)	37.81 (3.49)
BMI Category			
BMI < 25	53 (34%)	40 (36%)	13 (28%)
BMI >= 25	104 (66%)	70 (64%)	34 (72%)

[1]Mean (SD); n (%)

For *SIRT3*, the median expression values were also similar across the four groups, although the BMI 25 or higher groups showed slightly higher median values.

Overall, all three genes showed overlapping expression distributions across pregnancy status and BMI categories. The observed differences were relatively small, and these plots alone do not establish statistical significance.

Comparison of Histograms Across Three Genes

I compared the distributions of *CNR1*, *SOCS5*, and *SIRT3* gene expression across 157 pregnancy samples.

CNR1 had most expression values around 4.8 to 4.9. Its distribution was slightly right-skewed, with a few samples showing higher expression values.

SOCS5 had most expression values around 7.4 to 7.6. Its distribution was more balanced around the center and showed slightly greater variation.

SIRT3 had most expression values around 8.8 to 9.0. Its distribution was more concentrated around the center, with a few values at either end.

Overall, the three genes showed different expression distributions. *SOCS5* had the largest standard deviation (0.146), while *SIRT3* had the smallest (0.123).

Summary statistics by pregnancy status

The dataset included 157 pregnancy samples, with 110 normal pregnancies and 47 cases of preeclampsia. The preeclampsia group had a higher average maternal BMI (31.16 versus 27.99) and a lower average gestational age at delivery (37.81 versus 39.25 weeks). These are descriptive differences.

Heatmap of ten genes

The heatmap shows how expression varies across the samples. I scaled each gene by row and used clustering to group similar patterns. The colored bars identify pregnancy status and BMI category. I did not see a clear separation between pregnancy groups based on this plot alone.

Additional visualization: Parallel coordinates plot

I used a parallel coordinates plot to compare *CNR1*, *SOCS5*, and *SIRT3* across the two pregnancy groups. Each thin line shows a sample, and the thicker line shows the group average. The average profiles were fairly similar across groups. This plot alone does not show statistical significance.

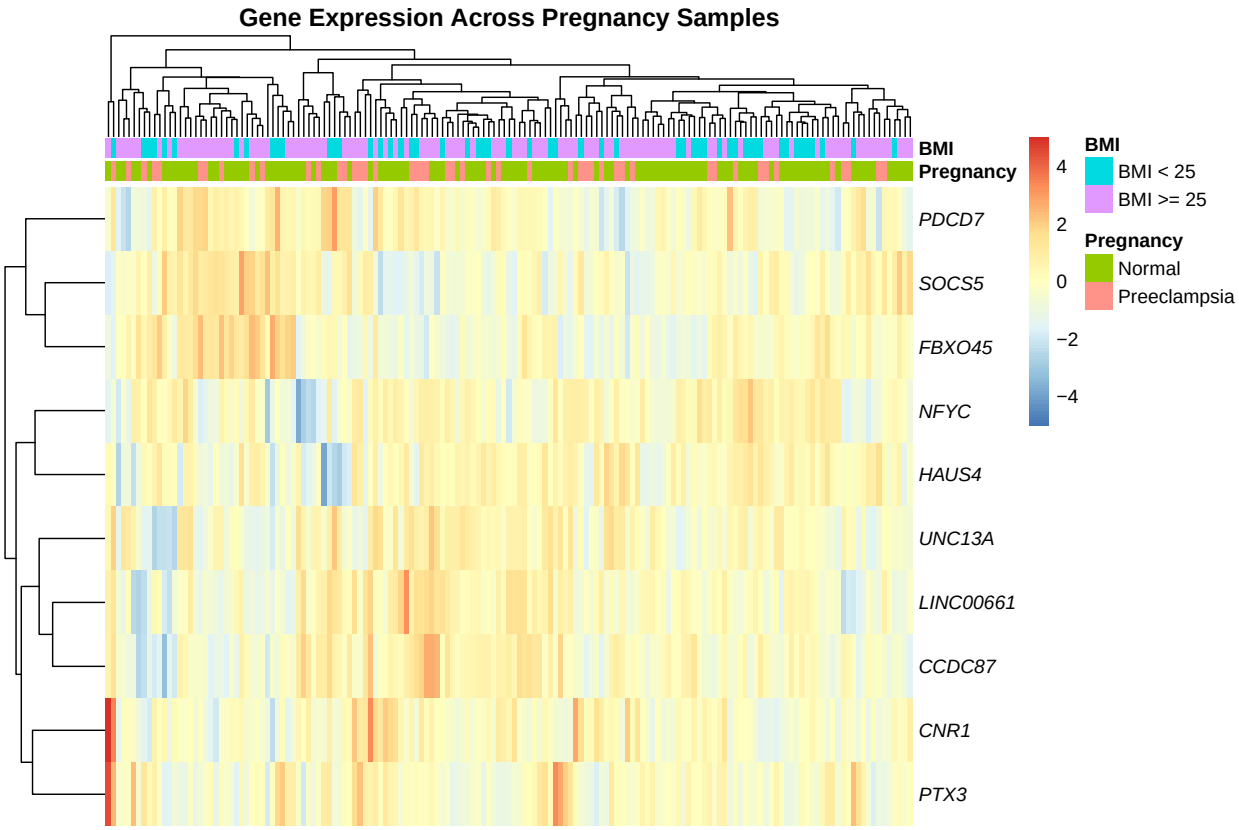


Figure 10: Heatmap of standardized expression for ten genes across 157 samples, with hierarchical clustering and pregnancy status and BMI category annotations.

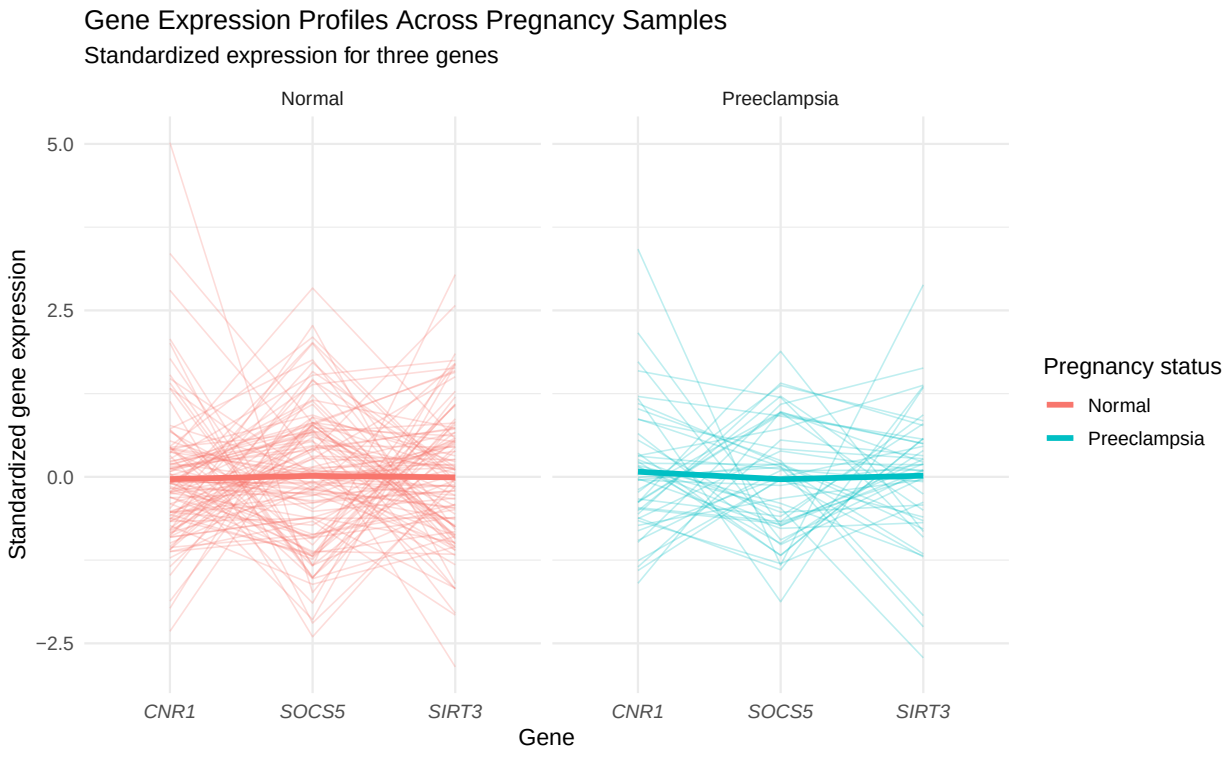


Figure 11: Parallel coordinates plot of standardized expression for three genes. Thin lines show individual samples and thick lines show the mean profile within each pregnancy group.